Rice-Husk Ash versus Recycled Concrete Fines: Why the Two Corpora Need Different Models

A side-by-side comparison of the two ≤ 60 MPa strength studies · September 2026

Abstract

Both studies predict the compressive strength of blended concrete from a literature corpus, restricted to samples of 60 MPa or below and validated by holding out whole papers. They agree on what makes the problem solvable and disagree on how to model it. Two ingredients transfer unchanged: the paper's own control strength, without which neither corpus is learnable, and the requirement that strength rise with curing age and with control strength. Two do not: RHA is best with three inputs and lightly regularised trees, while RCF needs the material descriptors and much larger leaves. A controlled swap of the two recipes shows each corpus is worse under the other's, and the cause is structural — RCF has twice the papers but 69% of its rows are repeat specimens, against 23% for RHA.

1. The two corpora

Table 1: The modelling sets after removing samples above 60 MPa (blend rows with a same-age control).

Property	RHA	RCF
Rows / papers	308 / 21	688 / 41
Rows per paper	14.7	16.8
Repeat specimens (rows sharing paper, level and age)	23%	**69%**
Spread of the target (SD)	13.7 MPa	14.1 MPa
Material	one ash, chemically similar across papers	powders, fines and blends of differing origin

2. What transfers

The control strength. On both corpora, predicting absolute strength without the paper's own 0%-replacement result is worse than predicting the corpus average (RHA $R^2 = 0.14$ at best, RCF -0.13); supplying it makes the problem learnable. It is the top-ranked input in both final models, and it is legitimate: the plain mix can always be cast and tested first.

Physics constraints. Requiring strength never to fall as curing age or control strength rise is the single most valuable modelling choice in both studies — worth about $+0.12$ R^2 on RHA and $+0.07$ on RCF.

What else behaves the same. Only tree ensembles work (Ridge and small neural nets fail badly on both); missing-value strategy barely matters; text embeddings of the papers never beat the structured columns; and grouped validation by paper is essential, since ordinary random splits would leak a paper's own control level into its test rows.

3. What differs, and by how much

The two recipes differ in exactly two places: which features they use, and how large a tree leaf must be. Holding everything else fixed — same splits, same monotone constraint, same learning rate and tree count — each recipe was run on both corpora.

Table 2: Cross-applying the recipes. Each corpus scores best under its own (green), and each factor can be read off separately.

Configuration	RHA R^2	RCF R^2
RHA recipe — base inputs only, small leaves (min. 5)	0.346 ± 0.249	0.268 ± 0.218
base inputs only, large leaves (min. 20)	0.328 ± 0.256	0.286 ± 0.212
base + categorical + fineness, small leaves	0.251 ± 0.281	0.279 ± 0.218
RCF recipe — base + categorical + fineness, large leaves	0.267 ± 0.269	0.309 ± 0.188

Reading the table by column:

- **Adding the material descriptors** costs RHA about 0.09 R^2 and gains RCF about 0.02–0.03. On RHA the extra columns are noise; on RCF they carry signal.
- **Enlarging the leaves** from 5 to 20 samples costs RHA 0.02 and gains RCF 0.02–0.03.
- **Together**, using the wrong corpus's recipe costs 0.079 R^2 on RHA and 0.041 on RCF.

4. Why the method changed for RCF

Reason 1: RCF has twice as many papers, so it can afford more inputs. Grouped validation means the effective sample size is the number of papers, not rows: 21 for RHA against 41 for RCF. With 21 papers, adding a dozen sparsely reported powder properties gives the tree room to split on a column that happens to separate three training papers and means nothing on a held-out one — the feature-set study for RHA showed every added group within noise of the three base inputs, and the tuned model kept only those three. With 41 papers, the same columns become usable: RCF's base set alone is its *worst* configuration, and the descriptors lift it by a margin larger than the gap between any two tree models. Particle size in particular carries real signal for RCF (third-ranked input in the final model), which is unsurprising: recycled fines vary in grinding and origin, whereas rice-husk ash is chemically similar from paper to paper, so its properties add little beyond what the control strength already encodes.

Reason 2: RCF's rows are mostly repeat specimens, so it needs heavier regularisation. 69% of RCF's rows share a (paper, replacement level, age) key with another row — several specimens of the same nominal mix, differing only by scatter the corpus does not explain — against 23% for RHA. A tree allowed to isolate three samples per leaf can fit that scatter exactly, and none of it transfers to a new paper. Forcing at least 20 samples per leaf blocks this. The effect is large enough to have looked like an algorithmic difference: HistGradientBoosting defaults to 20 samples per leaf and XGBoost to about 3, which alone made HistGB appear the better library on RCF (0.269 vs 0.244). Equalising the setting reverses the ordering, and once both are tuned they finish level (0.308 vs 0.309). **The library never mattered; the leaf size did.**

5. Final models, side by side

Table 3: The tuned model each corpus settled on, scored by repeated grouped hold-out (40 repeats, 30% of papers held out).

	RHA	RCF
Model	XGBoost, depth 2	shallow boosted trees, depth 2 (HistGB or XGBoost, interchangeable)
Inputs	3: replacement, age, control	18: + categorical descriptors, fineness
Minimum samples per leaf	5	20
Monotone in age and control	yes	yes
R^2	0.35 ± 0.25	0.31 ± 0.19
RMSE / MAE (MPa)	10.9 / 8.2	11.0 / 8.4

The achievable accuracy is nearly identical — about 8–8.5 MPa of typical error on an unseen paper — despite the different routes. Neither corpus supports predicting the optimal replacement level: both optimum predictors lose to simply quoting the corpus-average optimum, and on RCF the target is additionally ill-posed, since in 66% of papers the strongest mix measured is the plain control.

6. The LLM comparison on each corpus

Table 4: gpt-oss:120b against the tuned model on each corpus's fixed held-out papers. The two rows are not equally strict — see the note.

Corpus (held-out set)	Tuned ML	gpt-oss (median of runs)	Inputs matched?
RHA — 6 papers, 49 points	0.778	0.703	no: the LLM also saw the paper id
RCF — 6 papers, 59 points	0.789–0.807	0.602	yes: strictly enforced

A caveat that applies to both studies. The RHA comparison, and an earlier RCF one, placed the paper identifier in the LLM's prompt while the tree model never received it. That column carries no strength information, but it tells the model which queries belong to the same paper, so it can anchor each block on that paper's control strength — something a tree cannot exploit. On RCF, measured directly, the identifier alone moves the LLM from R^2 0.204 to 0.668 per run. The RCF figure above therefore withholds it from both sides; the RHA figure does not, and should be read as favourable to the LLM.

On both corpora a single LLM run is unreliable at temperature 0 — repeated identical prompts give R^2 anywhere from −0.62 to 0.85 on RCF — so the median of five or more runs is the only usable form, and even then the tuned tree is ahead on strictly equal inputs.

7. Practical guidance

1. **Always supply the paper's own control strength** and constrain strength to rise with age and control. These two choices transfer across corpora and matter more than anything else.
2. **Let the corpus decide the feature set.** Count papers, not rows: below roughly 25 papers, keep to the three base inputs; above roughly 40, test the material descriptors, because they start to pay.
3. **Set the minimum leaf size from the share of repeat specimens**, not from library defaults. Where most rows are repeats, require 20 or more samples per leaf.
4. **Do not port a tuned recipe between corpora unchanged** — it costs 0.04–0.08 R^2 here. Re-run the ablation; it is inexpensive.
5. **Use the LLM as a fallback, not the primary tool**, and if you do compare the two, give both sides exactly the same columns.

Sources

- RHA: `rha_report.pdf` Part III; pipeline in `pipeline/13_ml_le60.py`, results in `pipeline/out_le60/`.
- RCF: `rcf_le60_report.pdf`; pipeline in `pipeline_rcf/09_ml_le60.py` and `pipeline_rcf/10_llm_le60.py`, results in `pipeline_rcf/out_le60/` and `out_llm_le60/`.
- Table 2 comes from a controlled cross-application run for this note: both corpora, the same 40 grouped splits, monotone constraints fixed, varying only feature set and minimum leaf size.